

Parallel Scenario

What is the average water consumption in liters of the top 3 S&P 500 companies in 2023, considering only UK branches?

Extract one value from each non-relational table, then aggregate

Company X			Company Y			Company Z			
2023 indicators per employee	Branch location		Branch locations 2023	Weekly liters (per employee)		Weekly usage per employee		Rbnach citise	
	UK	US		Water	Oil			Lodnon (UK)	Tokoy (Jaapn)
Weekly water usage 2022	5.89 gallons	7.16 gallons	France	22	31	Waetr	2022	0.022 m ³	0.031 m ³
Weekly water usage 2023	5.60 gallons	--	UK	21.5	not available		2023	0.024 m ³	0.028 m ³
Perturbations: Pivot + unit change + n.a.			Perturbations: Pivot + unit in header + n.a.			Perturbations: Row/col pivot + unit change + OCR			

Conversion: 5.6 gallons → 21.2 liters, 0.024 m³ → 24 liters → Answer: 22.23 liters
avg(21.2, 21.5, 24)

Sequential Scenario

What is the average 2023 investment amount of the main lab of the company where John Doe is CEO?

Restrict the calculation to investments done in AI and the financial sector

Follow implicit lookup keys across perturbed tables

CEO names			Facility assignment				R&D expenses			
Year	Compaines		Facilities	Compaines			Costs (2023)	Investments (millions)		
	X	Y		X	Y			AI Research	Finance Research	
2022	John Doe	Lee Seo-yeon	Primary	Atlanta Lab	Seoul Lab		Net cost	Atlanta Lab	42.1	38.9
2023	Jnae Mya	n.a.	Backup	Denver Hub	Busan Hub			Seoul Lab	53.8 - as written in [1]	n.a.
Perturbations: Pivot + n.a. + OCR			Perturbations: Pivot + empty column				Perturbations: Row/Col pivot + text insertion			

propagate values
Company X → Atlanta Lab → avg(42.1, 38.9) → Answer: 40.5